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Logarithmic Proper-Time Interference in Zeta-like Quantum Chaotic Spectra

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Abstract

We introduce a logarithmic proper-time observable for zeta-like quantum chaotic spectra. Writing a nontrivial zero as $\rho_k = \sigma_k + i\gamma_k$, the finite numerical transform uses the critical-line amplitude $\sigma = 1/2$ and assigns to the ordinate γ_k the level-dependent clock $\tau_k = \pi \log\log k/\log k$. The resulting ordered amplitude is not a spacing statistic but a phase-interference observable. The Riemann–von Mangoldt inverse-counting approximation, evaluated through its Lambert W inversion, is used as a smooth semiclassical background; throughout the paper this Lambert-based inverse approximation is denoted simply by the RVM background. The main point is that the local one-step residual vector gaps are Rayleigh-like, whereas the ordered residual window sums exhibit coherent finite blocks. The mechanism is interpreted through the classical fluctuation term $S(T)$ in the Riemann–von Mangoldt formula: the zeta–background ordinate defect is a reparametrization of the zero-counting fluctuation, and the logarithmic clock converts that fluctuation into an ordered phase modulation. In this sense, the paper introduces a complementary spacing-sensitive observable: it is consistent with the usual local GUE and Riemann–von Mangoldt picture, but extends the characterization of the zero sequence to mesoscopic and macroscopic phase organization that standard spacing statistics and zero-counting asymptotics do not display. A separate counting-coordinate construct is also recorded for the real coordinate: it is defined independently of RH and, by construction, vanishes on the critical coordinate while assigning a signed response to a supplied zero with real part different from 1/2. It is presented modestly as a formal real-coordinate bookkeeping device and as a possible starting point for deeper analytic refinements.

Keywords

Riemann Zeta Function; Quantum Chaos; GUE Statistics; Spectral Interference; Logarithmic Proper Time; Riemann–von Mangoldt Formula; Rayleigh Law; Block Coherence.

1. Introduction

One of the major advances in the modern understanding of the Riemann zeta function came from an unexpected connection between the physics of chaotic quantum systems and the mathematics of zeta zeros. Montgomery's pair-correlation work showed, under the Riemann Hypothesis and for a restricted test-function range, a two-point correlation law for the zeta-zero ordinates [5]. As recorded in the Montgomery–Odlyzko random-matrix narrative, Freeman Dyson recognized the same limiting form as the two-point correlation law of the Gaussian Unitary Ensemble of random matrix theory, the ensemble that appears in quantum systems without time-reversal symmetry [5, 6, 8]. Subsequent large-scale computations by Odlyzko gave strong numerical evidence for this Montgomery–Odlyzko/GUE picture [6]. This connection is now part of the standard interface between zeta zeros, random matrices, and quantum chaos; see also [9–14].

This physical background explains why a paper about zeta zeros may legitimately introduce spectral language. The usual random-matrix description tests the sequence through unfolded spacings, pair correlations, number variance, and related observables. These statistics examine the distribution of levels. They do not directly test how the ordered spectrum interferes after a nonlinear phase projection.

We write a nontrivial zero as

$$\rho_k = \sigma_k + i\gamma_k, \tag{1.1}$$

where the ordinates are ordered increasingly. Under the Riemann Hypothesis one has

$$\sigma_k = \frac{1}{2} \tag{1.2}$$

for every nontrivial zero. The computations in this article use tabulated critical-line ordinates and therefore evaluate the logarithmic proper-time transform at the critical-line amplitude parameter

$$\sigma = \frac{1}{2}. \tag{1.3}$$

We then use the spectral notation

$$E_k = \gamma_k. \tag{1.4}$$

The observable studied here assigns to the k-th level a logarithmic proper time

$$\tau_k = \pi \frac{\log\log k}{\log k}, \qquad k \geq 3, \tag{1.5}$$

and forms the complex term

$$z_k^{\zeta} = \exp\left(\frac{\tau_k}{2} + iE_k\tau_k\right). \tag{1.6}$$

The primary object is the ordered cumulative amplitude

$$S_N^{\zeta} = \sum_{k=3}^{N} z_k^{\zeta}. \tag{1.7}$$

The word ordered is essential. The observable is not the unordered cloud of the points z_k^{ζ}. It is the cumulative interference produced by the spectral sequence in its natural order.

The smooth comparison sequence is obtained from the inverse Riemann–von Mangoldt counting approximation. In the computations this inverse-counting background is evaluated by the corresponding Lambert W inversion of the mean zero-counting law. From this point on, this Lambert-based inverse Riemann–von Mangoldt approximation is denoted simply by RVM. We denote the corresponding background ordinates by

$$E_k^{(0)} = \gamma_k^{\mathrm{RVM}}. \tag{1.8}$$

This RVM sequence is used as a semiclassical control: it supplies the smooth Weyl-type spectral skeleton against which the true zeta fluctuation is measured.

The main finite observation is that the individual transformed zeta and RVM terms may be close at the term-cloud scale, whereas their ordered cumulative sums are sharply different. The residual sequence is locally simple but globally ordered: its one-step vector gaps are Rayleigh-like, while its ordered window sums exhibit large coherent excess.

From the mathematical side, the point of the construction is that it is an exponential sum over zeta-zero ordinates which reveals a complex ordered morphology and a strong separation between the smooth RVM background and the true zeta response. Standard exponential sums over zeros, explicit-formula sums, fractional-part statistics, and random-matrix spacing observables form a large literature. Within the literature cited here, and to the author's knowledge, no comparable exponential-sum observable has been isolated in which the same deterministic weight produces a small smooth RVM control while the zeta transform develops the observed finite two-nucleus, arm, and tube morphology. The claim is not that the effect is already an asymptotic theorem. The claim is that this finite morphology motivates a class of ordered phase-interference observables whose block behavior is not resolved by spacing diagnostics alone.

This viewpoint is not limited to zeta zeros. Any system with an ordered quantum chaotic spectrum, a smooth semiclassical background, and a physically meaningful level-dependent clock can be tested by the same construction. The article is therefore also a model for underexplored spectral approaches in which one studies not only level statistics, but ordered proper-time interference.

Remark 1.1. The physical systems motivating the language are described in Section 7. The relevant examples are quantum chaotic spectra, random-matrix spectra, trace-like spectral amplitudes, and effective scale geometries. The role of zeta in this article is spectral: the zero ordinates are used as a model quantum chaotic spectrum.

2. Standard background and status of the present observable

Remark 2.1. The Riemann–von Mangoldt formula, the interpretation of γ_k as zero ordinates, and the use of unfolded gaps are standard material in analytic number theory; see [2–4]. The relation between zero statistics and random matrices is literature-based material; see Montgomery and Odlyzko [5, 6], Katz–Sarnak and Rudnick–Sarnak [11, 12], and Keating–Snaith and Conrey–Snaith [13, 14]. The logarithmic proper-time observable defined below is the original object studied in this article.

Definition 2.2. The classical locally unfolded zeta gap is a quantity of the form

$$s_k^{\mathrm{cl}} = \frac{\gamma_{k+1} - \gamma_k}{\langle \gamma_{k+1} - \gamma_k \rangle_{\mathrm{loc}}}. \tag{2.1}$$

It is a nearest-neighbor spacing statistic.

Remark 2.3. The observable in (1.7) is not the statistic in (2.1). It tests the ordered phase field

$$\theta_k^{\zeta} = E_k \tau_k, \tag{2.2}$$

with τ_k as in (1.5). Consequently, agreement with GUE at the level of local spacings does not force the logarithmic proper-time observable to be locally GUE-distributed.

3. Position relative to known exponential sums

Remark 3.1. Standard exponential sums over zeros enter the explicit formula and related Landau–Gonek type formulae; see [2, 4]. Other phase statistics, such as fractional parts of linear forms in zero ordinates, have been studied in connection with pair correlation and prime distribution questions; see [7]. Random-matrix observables test unfolded local spectral statistics; see [5, 6, 8].

Remark 3.2. The observable introduced here differs from the standard objects in Remark 3.1. It is not a fixed-parameter sum of the form $\sum e^{i\gamma t}$, not an explicit-formula sum over x^{ρ}, not a fractional-part statistic of $\alpha\gamma$, and not an unfolded spacing statistic. It is an ordered cumulative sum of the phase field $\gamma_k\tau_k$, with $\tau_k = \pi \log\log k / \log k$, compared against the same transform applied to the RVM background. Within the cited literature, no comparable publication is known to the author in which a single exponential-sum transform reveals a morphology of the present complexity while sharply separating the RVM background from the zeta response. This motivates treating (1.7) as the first member of a broader family of ordered zeta observables.

Heuristic 3.3. This is a heuristic interpretation. The observed lower nucleus, arm, upper entry, and tube are not arbitrary plotting artifacts. They indicate that the logarithmic clock imposes a susceptibility skeleton in k, and that the zeta ordinates excite this skeleton through ordered phase correlations. The visible structures therefore suggest an underlying law governing block coherence in the phase field $\gamma_k\tau_k$. Standard GUE spacing statistics are not designed to detect this law, because they discard the ordered proper-time interference encoded in S_N^{ζ} and $B_W^{\zeta}(k)$.

4. Logarithmic proper-time transform

Definition 4.1. For $k \geq 3$, define the logarithmic proper time

$$\tau_k = \pi \frac{\log\log k}{\log k}. \tag{4.1}$$

The lower bound $k \geq 3$ is only a domain convention: it ensures that $\log\log k$ is real and positive. It has no independent spectral meaning.

Definition 4.2. Let $E_k = \gamma_k$, and let σ be a real amplitude parameter. The zeta logarithmic proper-time term is

$$z_k^{\zeta}(\sigma) = \exp\left((\sigma + iE_k)\,\tau_k\right). \tag{4.2}$$

The finite computations in this article use

$$z_k^{\zeta} = z_k^{\zeta}\left(\frac{1}{2}\right) = e^{\tau_k/2}e^{iE_k\tau_k}. \tag{4.3}$$

Definition 4.3. Let $E_k^{(0)} = \gamma_k^{\mathrm{RVM}}$ denote the Lambert-based RVM background ordinate. For a real amplitude parameter σ, the background term is

$$z_k^{\mathrm{RVM}}(\sigma) = \exp\left(\left(\sigma + iE_k^{(0)}\right)\tau_k\right). \tag{4.4}$$

The finite computations in this article use

$$z_k^{\mathrm{RVM}} = z_k^{\mathrm{RVM}}\left(\frac{1}{2}\right). \tag{4.5}$$

Definition 4.4. The residual transformed term is

$$d_k = z_k^{\zeta} - z_k^{\mathrm{RVM}}. \tag{4.6}$$

Definition 4.5. The cumulative amplitudes are

$$S_N^{\zeta} = \sum_{k=3}^{N} z_k^{\zeta}, \tag{4.7}$$

$$S_N^{\mathrm{RVM}} = \sum_{k=3}^{N} z_k^{\mathrm{RVM}}, \tag{4.8}$$

$$S_N^{\mathrm{diff}} = \sum_{k=3}^{N} d_k. \tag{4.9}$$

Remark 4.6. By the definition of d_k, the finite cumulative sums satisfy the algebraic decomposition

$$S_N^{\zeta} = S_N^{\mathrm{RVM}} + S_N^{\mathrm{diff}}. \tag{4.10}$$

This identity is bookkeeping, not an explanatory theorem. The curve under study is S_N^{ζ}. RVM is a diagnostic control used to expose which features are imposed by the logarithmic clock and which features are strongly excited by the true ordinates.

5. Spectral proper-time interpretation

Heuristic 5.1. This is a heuristic interpretation. Under the spectral identification (1.4), the phase of the transformed term is

$$\theta_k^{\zeta} = E_k\tau_k. \tag{5.1}$$

Thus (4.3) may be read as a spectral amplitude evaluated in a non-uniform proper-time gauge. The factor $e^{\tau_k/2}$ is a slowly varying amplitude, while the phase $E_k\tau_k$ carries the main interference structure.

Remark 5.2. A standard spectral trace or return amplitude would use a common time variable and terms of the form e^{iE_kt}. The present observable uses the level-dependent time τ_k. Therefore it tests ordered phase interference rather than only the distribution of spacings.

6. Semiclassical background and residual phase response

Definition 6.1. The spectral defect relative to the RVM background is

$$\delta E_k = E_k - E_k^{(0)}. \tag{6.1}$$

Remark 6.2. The residual has the exact factorization

$$d_k = z_k^{\mathrm{RVM}} \left(e^{i\delta E_k \tau_k} - 1 \right). \tag{6.2}$$

This identity isolates the relevant fluctuation δE_k, but it is only algebraic. The analytical question is whether the ordered sequence of these phase defects has correlations beyond a chosen null model.

Heuristic 6.3. This is a heuristic argument. If the phase defect $\delta E_k \tau_k$ is small enough to linearize the exponential, then

$$e^{i\delta E_k \tau_k} - 1 \approx i\delta E_k \tau_k. \tag{6.3}$$

Combining (6.3) with (6.2) gives

$$d_k \approx i z_k^{\mathrm{RVM}} \delta E_k \tau_k. \tag{6.4}$$

This gives the physical interpretation of the residual as a first-order phase response of the logarithmic proper-time observable to the spectral defect δE_k.

Remark 6.4. The spectral defect $\delta E_k = E_k - E_k^{(0)}$ is not a new source of randomness. It should be compared with the classical oscillatory term in the Riemann–von Mangoldt counting formula. If

$$N(T) = N_0(T) + S(T) + O(T^{-1}), \tag{6.5}$$

where

$$S(T) = \frac{1}{\pi} \arg \zeta \left(\frac{1}{2} + iT \right), \tag{6.6}$$

then the smooth inverse ordinate $E_k^{(0)}$ is defined heuristically by $N_0(E_k^{(0)}) = k$, whereas the true ordinate satisfies $N(\gamma_k) = k$. Linearizing (6.5) gives

$$\delta E_k = \gamma_k - E_k^{(0)} \approx -\frac{S(E_k^{(0)})}{N_0'(E_k^{(0)})}. \tag{6.7}$$

Since

$$N_0'(T) = \frac{1}{2\pi} \log \frac{T}{2\pi}, \tag{6.8}$$

this gives the heuristic scale relation

$$\delta E_k \approx (-1) \frac{2\pi S(E_k^{(0)})}{\log(E_k^{(0)}/2\pi)}. \tag{6.9}$$

Thus the zeta–background ordinate defect is essentially the zero-counting fluctuation $S(T)$, rewritten as an ordinate displacement.

Heuristic 6.5. This is a heuristic mechanism. Selberg's central limit theorem implies Gaussian behaviour for the logarithm of the zeta function on the critical line; in particular, its imaginary part, and therefore the phase term underlying $S(T)$, has variance of order $\log \log T$, with normalization depending on the convention for the argument [17–19]. Combining this with (6.9) indicates that δE_k is a smoothed version of the Selberg-scale counting fluctuation. The logarithmic proper-time phase defect is therefore

$$\tau_k \delta E_k \approx (-1)\, \pi \frac{\log \log k}{\log k} \frac{2\pi S(E_k^{(0)})}{\log(E_k^{(0)}/2\pi)}. \tag{6.10}$$

This connects the observed residual phase modulation with established zeta fluctuations rather than with an unexplained new random source. The finite block coherence observed below is interpreted as the accumulation of these Selberg-scale phase defects under the slowly varying logarithmic clock.

7. Physical systems represented by the observable

Remark 7.1. This section records the physical link. The article does not assert that the zeta zeros are the measured spectrum of a known laboratory Hamiltonian. The claim is that the construction defines a spectral observable naturally expressed in the same language used for quantum chaotic spectra.

(i) **Quantum chaotic spectra.** The identification $E_k = \gamma_k$ places the zeta ordinates in the same formal role as energy levels. The BGS conjecture relates chaotic quantum spectra to random-matrix statistics [9]; the zeta zeros enter this circle through the Montgomery–Odlyzko/Dyson connection [5, 6, 14].

(ii) **Trace-like spectral amplitudes.** The sum S_N^ζ is analogous to a finite spectral phase amplitude, except that the time is τ_k rather than a common time t. This is why the observable is closer to ordered spectral interference than to a spacing histogram.

(iii) **Semiclassical background plus spectral defect.** The RVM sequence is treated as a smooth Weyl-type background. The residual d_k is the phase response to the fluctuation δE_k, as shown in Remark 6.2 and Heuristic 6.3.

(iv) **Effective scale geometry.** Section 14 rewrites τ_k as a redshift factor in the scale coordinate $u = \log k$. This gives a formal link with proper-time and redshift language, without assuming a physical spacetime metric.

8. Numerical protocol and limitations

Remark 8.1. The numerical study uses the first 2,001,052 zeta-zero ordinates from Odlyzko's public zero tables, distributed in the commonly used `zeros6` data set and documented as accurate to about $4 \cdot 10^{-9}$ in the ordinate values [15, 16]. The smooth comparison sequence is the corresponding Lambert-based RVM inverse-counting background constructed from the same index range. No private filename is part of the mathematical definition. A reproducible implementation needs only the public zero ordinates, the background construction, the logarithmic clock τ_k, and the figure-generation code.

Warning 8.2. Maxima such as $\max_k R_W^{\mathrm{diff}}(k)$ are selected after scanning many overlapping windows. They are therefore subject to a look-elsewhere effect. The values reported here show the size and location of the observed windows; they are not p-values. A significance claim would require a null ensemble, for example locally matched random phases, shuffled residual blocks, synthetic $S(T)$-scale perturbations of the RVM background, or random-matrix spectra passed through the same logarithmic proper-time transform.

Remark 8.3. Several numerical values are printed with many digits in order to identify exact output from the computation. Those digits should not be interpreted as statistical precision. In particular, the near-zero exponent in the Rayleigh comparison should ultimately be accompanied by confidence intervals or stability tests over the fitted ε-range.

9. Finite cumulative geometry

Definition 9.1. For the finite curve S_N^ζ, we use the descriptive names lower nucleus, arm, upper entry, and upper tube for the visible regions shown in Figure 2.

Warning 9.2. The names in Definition 9.1 are finite descriptive labels. They are not asymptotic theorems.

Experiment 9.3. For $N = 2000000$, the finite computation gives

$$S_N^\zeta = 108.3327 + 133.6213i, \tag{9.1}$$

$$S_N^{\mathrm{RVM}} = 1.1510 - 4.4564i, \tag{9.2}$$

$$S_N^{\mathrm{diff}} = 107.1817 + 138.0777i. \tag{9.3}$$

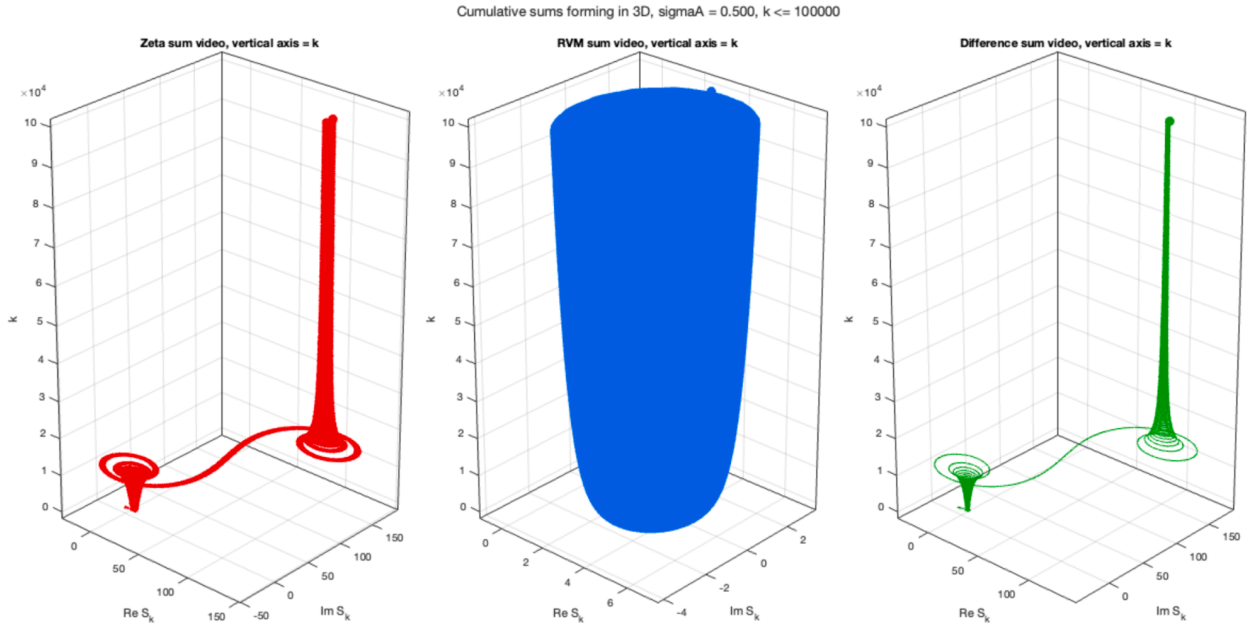


Figure 1: Ordered cumulative sums for $k \leq 100000$. The zeta curve is the primary object; the RVM curve is a smooth control; the difference curve isolates the residual contribution.

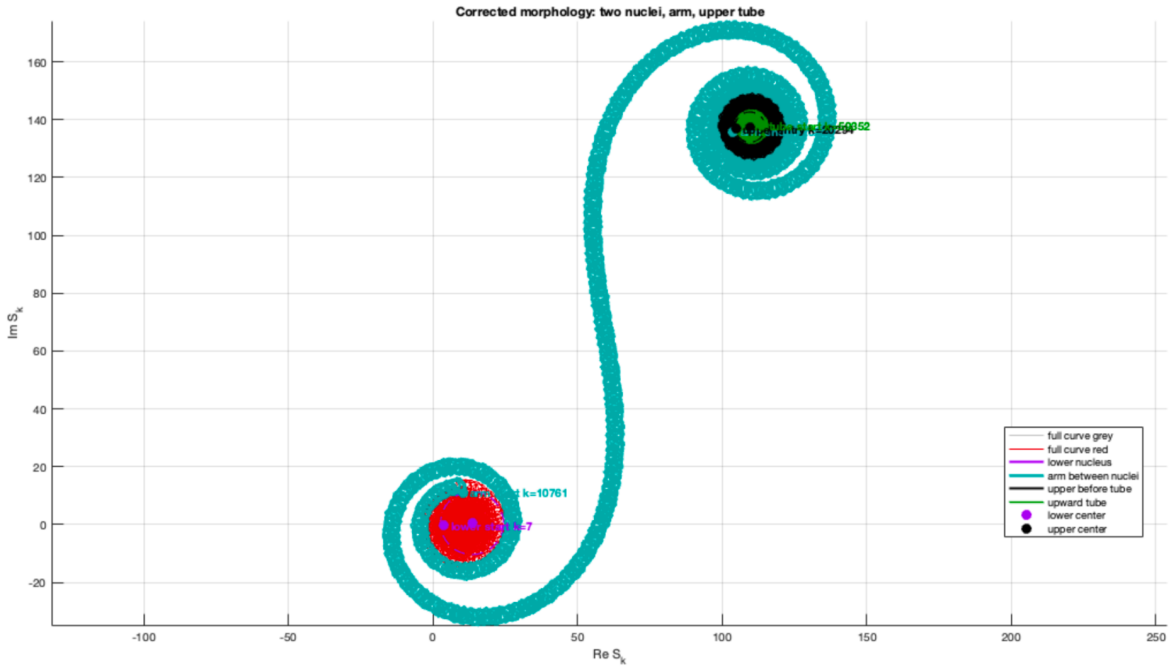


Figure 2: Two-dimensional morphology of the zeta cumulative curve. The labels are descriptive finite-regime names: lower nucleus, arm, upper entry, and upper tube.

The corresponding moduli are

$$|S_N^{\zeta}| = 172.0192, \tag{9.4}$$
$$|S_N^{\mathrm{RVM}}| = 4.6027, \tag{9.5}$$
$$|S_N^{\mathrm{diff}}| = 174.7952. \tag{9.6}$$

The final modulus ratios are

$$\frac{|S_N^{\mathrm{RVM}}|}{|S_N^{\zeta}|} = 0.02676, \tag{9.7}$$
$$\frac{|S_N^{\mathrm{diff}}|}{|S_N^{\zeta}|} = 1.01614. \tag{9.8}$$

10. Ordered window amplitudes

Definition 10.1. For $x \in \{\zeta, \mathrm{RVM}, \mathrm{diff}\}$, define the window amplitude

$$B_W^x(k) = \sum_{j=k}^{k+W-1} z_j^x. \tag{10.1}$$

For $x = \mathrm{diff}$, this means

$$B_W^{\mathrm{diff}}(k) = \sum_{j=k}^{k+W-1} d_j. \tag{10.2}$$

Definition 10.2. The random-walk normalized window excess is defined by

$$R_W^x(k) = \frac{|B_W^x(k)|}{\sqrt{W}\,\mathrm{rms}_{j\in[k,k+W-1]}(z_j^x)}. \tag{10.3}$$

Definition 10.3. The window cancellation statistic is defined by

$$C_W^x(k) = 1 - \frac{|B_W^x(k)|}{\sum_{j=k}^{k+W-1} |z_j^x|}. \tag{10.4}$$

Remark 10.4. Values $R_W^x(k)$ of order one are compatible with a random-walk scale. Larger values indicate stronger coherent alignment inside the ordered block. Values $C_W^x(k)$ close to one indicate strong cancellation, while smaller values indicate stronger internal alignment.

Experiment 10.5. For the first $N = 2000000$ ordinates used in the computation, the residual window excess gives

$$\max_k R_{500}^{\mathrm{diff}}(k) = 8.4750, \tag{10.5}$$
$$\max_k R_{1000}^{\mathrm{diff}}(k) = 11.8480, \tag{10.6}$$
$$\max_k R_{2000}^{\mathrm{diff}}(k) = 15.8891, \tag{10.7}$$
$$\max_k R_{5000}^{\mathrm{diff}}(k) = 11.0871. \tag{10.8}$$

The corresponding windows lie in the finite transition region near $k = 1.3 \cdot 10^4$ to $1.5 \cdot 10^4$.

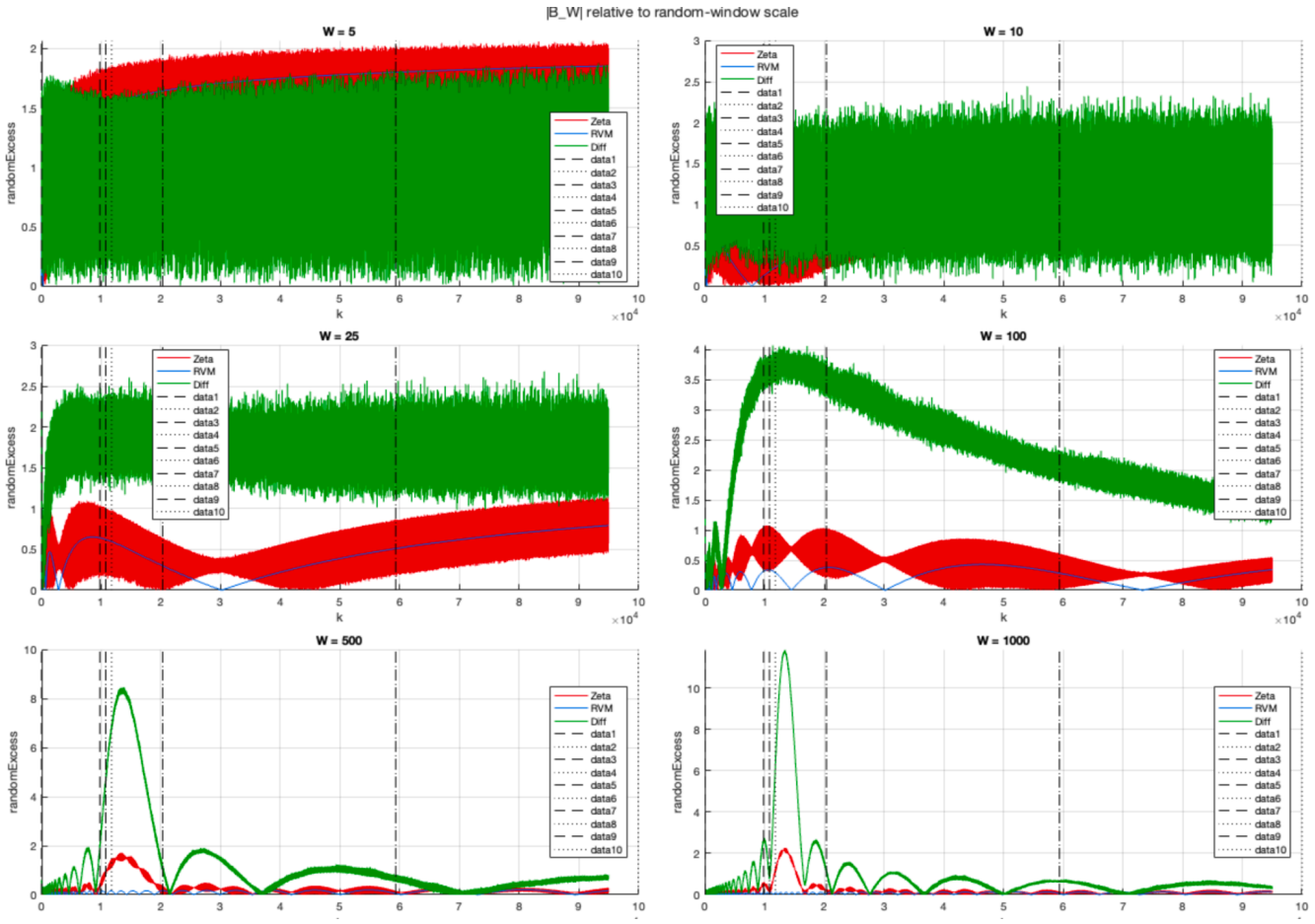


Figure 3: Local window diagnostics for the logarithmic proper-time transform. The panels compare zeta, RVM, and residual window behaviour. The key point is that RVM is not flat: it shows the smooth susceptibility skeleton imposed by τ_k, while the zeta and residual signals excite that skeleton more strongly.

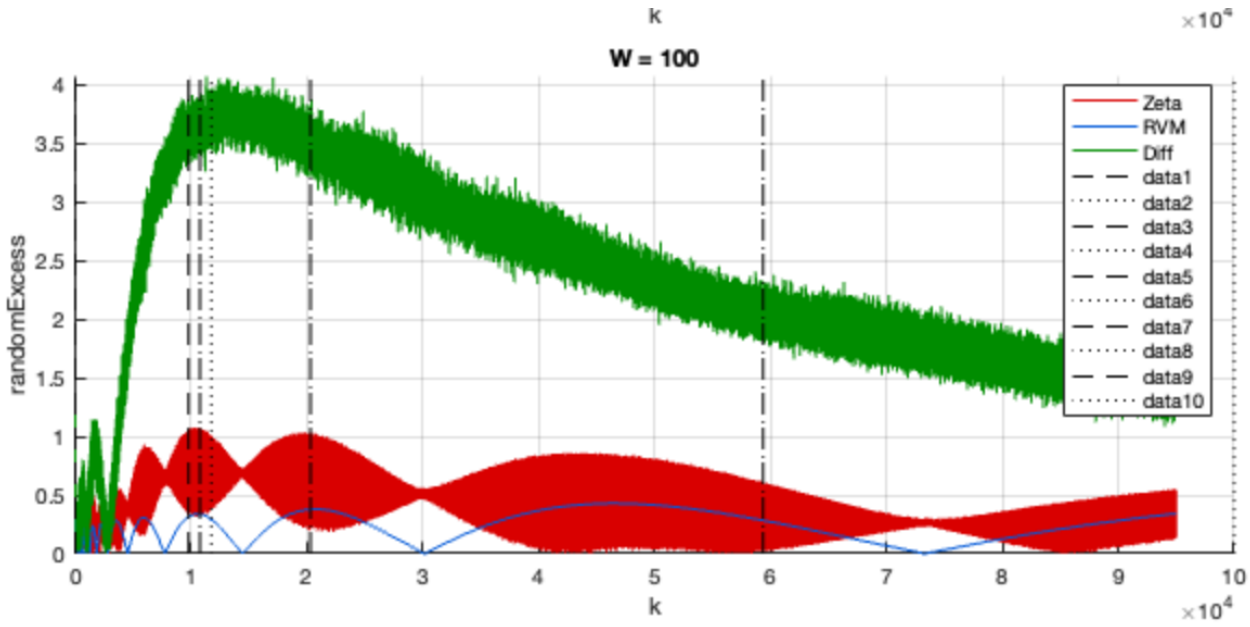


Figure 4: The $W = 100$ window panel. This figure resolves the early finite region where the arm begins to form. The residual and zeta curves show stronger excitation than the RVM control.

Experiment 10.6. For comparison, the RVM background gives

$$\max_k R^{\mathrm{RVM}}_{500}(k) = 0.3040, \tag{10.9}$$

$$\max_k R^{\mathrm{RVM}}_{1000}(k) = 0.2172, \tag{10.10}$$

$$\max_k R^{\mathrm{RVM}}_{2000}(k) = 0.1561, \tag{10.11}$$

$$\max_k R^{\mathrm{RVM}}_{5000}(k) = 0.09884. \tag{10.12}$$

Heuristic 10.7. This is a heuristic argument. For any finite sequence z_j, one has

$$|B_W(k)|^2 = \sum_{a=0}^{W-1}\sum_{b=0}^{W-1} z_{k+a}\overline{z_{k+b}}. \tag{10.13}$$

Separating diagonal and off-diagonal terms gives

$$|B_W(k)|^2 = \sum_{a=0}^{W-1} |z_{k+a}|^2 + 2\,\mathrm{Re}\sum_{0\le a<b\le W-1} z_{k+a}\overline{z_{k+b}}. \tag{10.14}$$

The diagonal part sets the random-walk scale. The off-diagonal part determines whether the ordered block is coherently enhanced or suppressed. In the present transform, the phase differences are

$$\theta^{\zeta}_{k+a} - \theta^{\zeta}_{k+b} = \gamma_{k+a}\tau_{k+a} - \gamma_{k+b}\tau_{k+b}. \tag{10.15}$$

Thus the window excess probes ordered phase correlations of the logarithmic proper-time field.

10.1. Robustness: Null test with synthetic Selberg-scale noise

Question 10.8. Does independent Gaussian noise of the same scale as the measured spectral defect reproduce the controlled zeta morphology, or does it over-excite the logarithmic transform?

The answer is tested by a null model: inject independent Gaussian noise with the same mean and standard deviation as the measured spectral defect $\delta E_k = \gamma_k - \gamma_k^{\mathrm{RVM}}$ into the smooth RVM sequence, apply the same logarithmic proper-time transform, and compare both window amplitudes and final cumulative amplitudes.

Definition 10.9. For the RVM ordinates γ_k^{RVM}, construct a synthetic perturbed sequence

$$\gamma_k^{\mathrm{noisy}} = \gamma_k^{\mathrm{RVM}} + \varepsilon_k, \quad \varepsilon_k \sim \mathcal{N}(\mu_\delta, \sigma_\delta^2), \tag{10.16}$$

where $\mu_\delta = \mathrm{mean}(\gamma_k - \gamma_k^{\mathrm{RVM}})$ and $\sigma_\delta^2 = \mathrm{var}(\gamma_k - \gamma_k^{\mathrm{RVM}})$ are the empirical mean and variance of the true spectral defect.

Experiment 10.10. For the null-test phase-walk visualization, $N = 800$ is used only to keep the three displayed paths readable. This should not be confused with the main numerical data set, which contains 2,001,052 Odlyzko zero ordinates. On this displayed segment, the spectral defect statistics are

$$\mu_\delta = -0.727324, \tag{10.17}$$

$$\sigma_\delta = 0.406515. \tag{10.18}$$

Apply the logarithmic proper-time transform to three sequences: (i) γ_k (true zeta), (ii) γ_k^{RVM} (pure RVM), and (iii) $\gamma_k^{\mathrm{noisy}}$ (RVM + synthetic Gaussian noise of scale $(\mu_\delta, \sigma_\delta)$).

Compute final cumulative amplitudes and window amplitudes:

$$|S_N^{\zeta}| = 13.004641, \tag{10.19}$$
$$|S_N^{\mathrm{RVM}}| = 5.263621, \tag{10.20}$$
$$|S_N^{\mathrm{noisy}}| = 27.426514. \tag{10.21}$$

For window amplitude $\max_k |B_W^x(k)|$ with $W = 500$:

$$\text{RVM pure} : 3.50, \tag{10.22}$$
$$\text{RVM} + \text{noise} : 27.78, \tag{10.23}$$
$$\text{Zeta real} : 11.67. \tag{10.24}$$

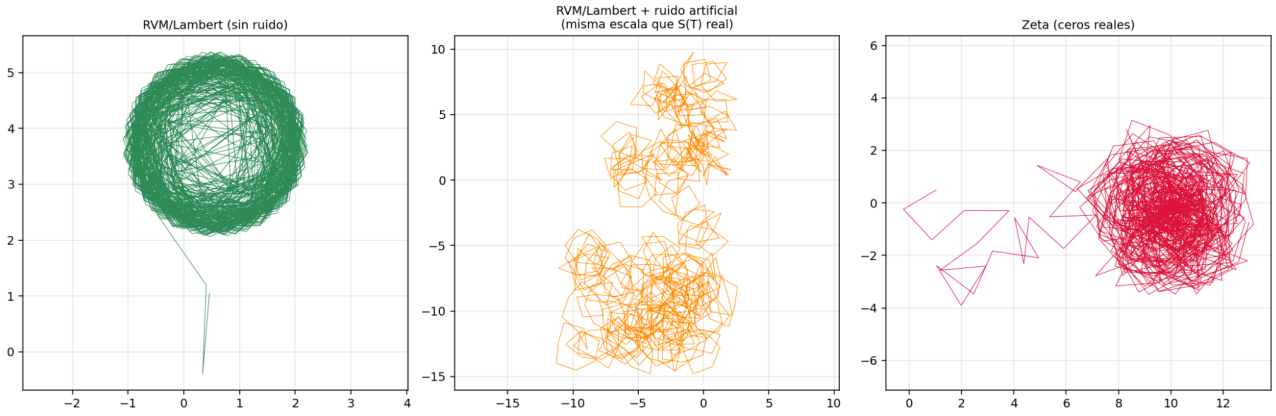


Figure 5: Null test for independent phase noise. Left: the pure RVM background is confined and smooth. Center: adding independent Gaussian noise of the same scale as the measured zeta–RVM spectral defect strongly over-excites the logarithmic proper-time transform. Right: the true zeta zeros produce an irregular but controlled response. For $W = 500$, the maximum window amplitudes are 3.50 for pure RVM, 27.78 for RVM plus independent noise, and 11.67 for the true zeta sequence. The final cumulative moduli are 5.26, 27.43, and 13.00, respectively. The null model therefore shows that unstructured phase noise can trigger the transform but overshoots the zeta response; the observed zeta spirals and finite windows are controlled by the ordered zeta phase defect rather than by independent noise alone.

Remark 10.11. The null test should be read in the opposite way from a naive noise explanation. Independent Gaussian perturbations of the correct one-step scale do not simply reproduce the zeta response. They over-excite the logarithmic transform. In the finite test reported above,

$$\max_k |B_{500}^{\mathrm{RVM}}(k)| = 3.50, \tag{10.25}$$
$$\max_k |B_{500}^{\mathrm{noise}}(k)| = 27.78, \tag{10.26}$$
$$\max_k |B_{500}^{\zeta}(k)| = 11.67, \tag{10.27}$$

and

$$|S_N^{\mathrm{RVM}}| = 5.26, \tag{10.28}$$
$$|S_N^{\mathrm{noise}}| = 27.43, \tag{10.29}$$
$$|S_N^{\zeta}| = 13.00. \tag{10.30}$$

Thus the logarithmic transform is sensitive to phase noise of the correct scale, but the zeta sequence is not behaving as independent Gaussian noise. The true zeta response is intermediate and controlled: much

larger than the smooth RVM background, but substantially smaller than the independently perturbed RVM sequence. This supports the interpretation that the visible spirals and finite window structures are produced by zeta-specific ordered correlations in the phase defect, not by unstructured noise alone.

Heuristic 10.12. This is a heuristic interpretation. The null model shows two separate effects. First, fluctuations of the correct scale are sufficient to activate the logarithmic clock. Second, if those fluctuations are made independent, the response is too large. The true zeta sequence therefore appears to combine Selberg-scale phase fluctuations with correlations that regulate the accumulated amplitude. In this sense the null test does not reduce the observed spirals to noise; it separates mere scale sensitivity from the ordered zeta correlations that control the final morphology.

11. Phase portraits and local drift

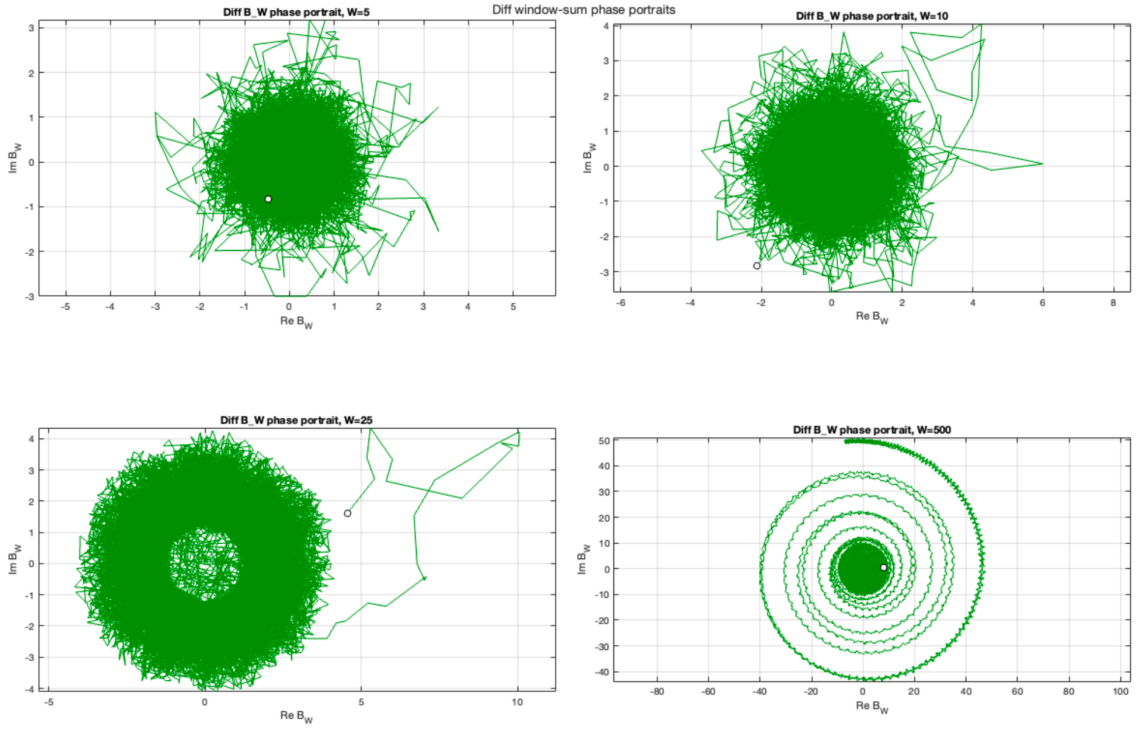


Figure 6: Phase portraits and local drift for selected window sums. The residual windows form organized loops rather than featureless scatter, and the local drift concentrates near the same finite transition region identified by the window diagnostics.

Remark 11.1. The phase portraits are not independent assumptions. They are local projections of the same ordered window amplitudes defined in (10.1). Their role is to show that the residual sequence is not featureless after ordered summation.

12. Nearest-neighbor gaps of the transformed function

Definition 12.1. The direct vector gaps of the transformed zeta and RVM terms are

$$D_k^{\zeta} = |z_{k+1}^{\zeta} - z_k^{\zeta}|, \tag{12.1}$$

$$D_k^{\mathrm{RVM}} = |z_{k+1}^{\mathrm{RVM}} - z_k^{\mathrm{RVM}}|. \tag{12.2}$$

The residual vector gap is

$$D_k^{\mathrm{diff}} = |d_{k+1} - d_k|. \tag{12.3}$$

Definition 12.2. The angular functional gaps are

$$\Delta\theta_k^{\zeta} = \gamma_{k+1}\tau_{k+1} - \gamma_k\tau_k, \tag{12.4}$$

$$\Delta\theta_k^{\mathrm{RVM}} = \gamma_{k+1}^{\mathrm{RVM}}\tau_{k+1} - \gamma_k^{\mathrm{RVM}}\tau_k, \tag{12.5}$$

$$\Delta\theta_k^{\mathrm{resid}} = \Delta\theta_k^{\zeta} - \Delta\theta_k^{\mathrm{RVM}}. \tag{12.6}$$

Definition 12.3. For a local unfolding window $W_{\mathrm{norm}} = 2000$, define

$$s_k^{\mathrm{func},\zeta} = \frac{\Delta\theta_k^{\zeta}}{\mathrm{movmedian}(\Delta\theta^{\zeta}, W_{\mathrm{norm}})}, \tag{12.7}$$

$$s_k^{\mathrm{func,RVM}} = \frac{\Delta\theta_k^{\mathrm{RVM}}}{\mathrm{movmedian}(\Delta\theta^{\mathrm{RVM}}, W_{\mathrm{norm}})}. \tag{12.8}$$

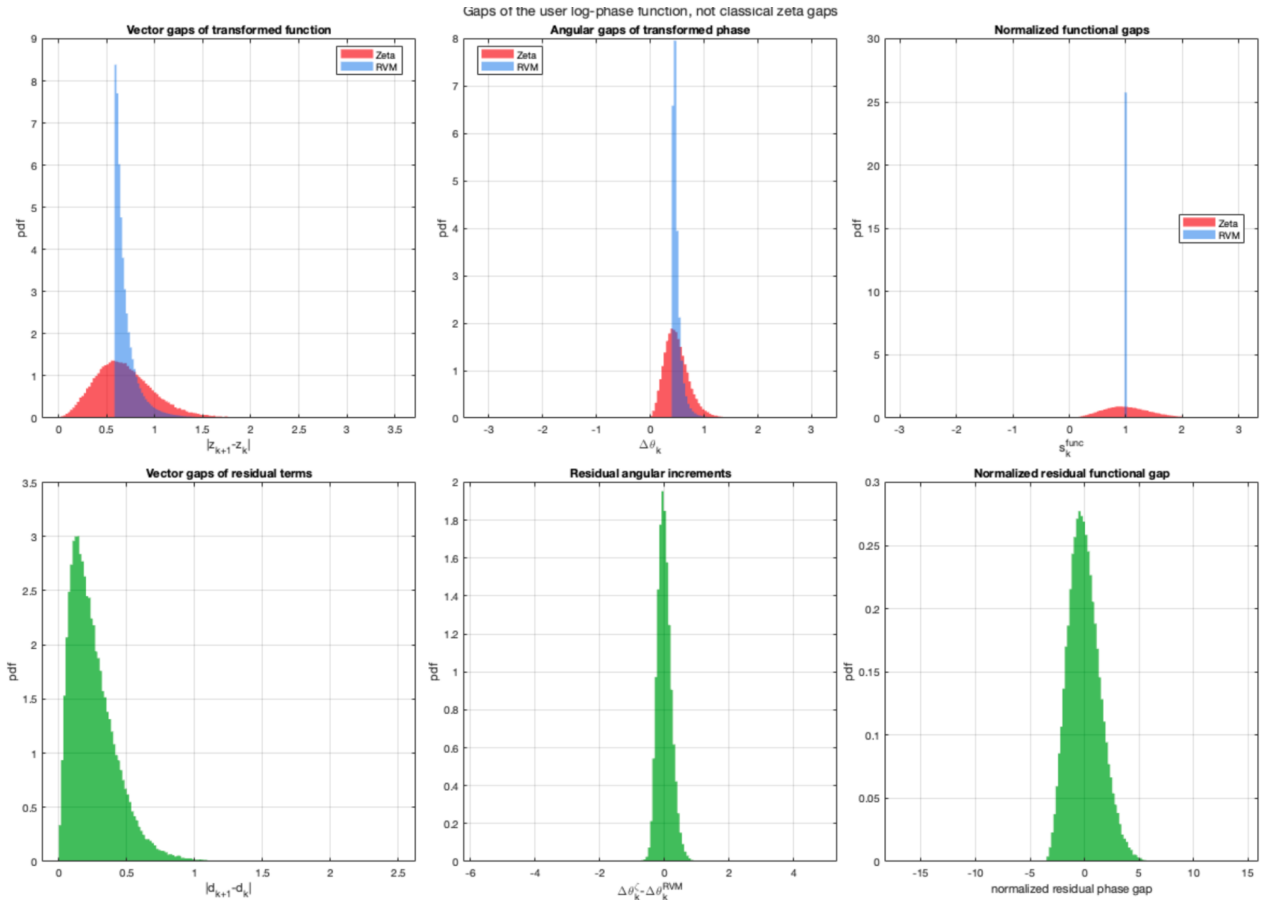


Figure 7: Nearest-neighbor gap diagnostics for the transformed function. The zeta functional gaps are much broader than the RVM control, while the residual one-step vector gap has its own radial law.

Experiment 12.4. For $N = 100000$ and $W_{\mathrm{norm}} = 2000$, the direct vector-gap statistics include

$$\mathrm{mean}(D^{\zeta}) = 0.6997, \tag{12.9}$$

$$\mathrm{mean}(D^{\mathrm{RVM}}) = 0.7048, \tag{12.10}$$

$$\mathrm{mean}(D^{\mathrm{diff}}) = 0.2569. \tag{12.11}$$

The standard deviations are

$$\mathrm{std}(D^{\zeta}) = 0.3374, \tag{12.12}$$

$$\mathrm{std}(D^{\mathrm{RVM}}) = 0.1720, \tag{12.13}$$

$$\mathrm{std}(D^{\mathrm{diff}}) = 0.1754. \tag{12.14}$$

For the locally normalized functional angular gaps,

$$\mathrm{std}(s^{\mathrm{func},\zeta}) = 0.4470, \tag{12.15}$$

$$\mathrm{std}(s^{\mathrm{func,RVM}}) = 0.04462. \tag{12.16}$$

Thus the zeta functional gaps have approximately ten times the standard deviation of the RVM control after local normalization.

13. Rayleigh local residual gaps

Definition 13.1. The mean-normalized residual vector gap is

$$s_k^{\mathrm{diffvec}} = \frac{|d_{k+1} - d_k|}{\mathrm{mean}_j |d_{j+1} - d_j|}. \tag{13.1}$$

Definition 13.2. The comparison laws used in the nearest-neighbor residual test are

$$p_{\mathrm{GUE}}(s) = \frac{32}{\pi^2} s^2 e^{-4s^2/\pi}, \tag{13.2}$$

$$p_{\mathrm{Poisson}}(s) = e^{-s}, \tag{13.3}$$

$$p_{\mathrm{Rayleigh}}(s) = \frac{s}{\sigma_R^2} \exp\left(-\frac{s^2}{2\sigma_R^2}\right), \qquad \sigma_R = \sqrt{\frac{2}{\pi}}. \tag{13.4}$$

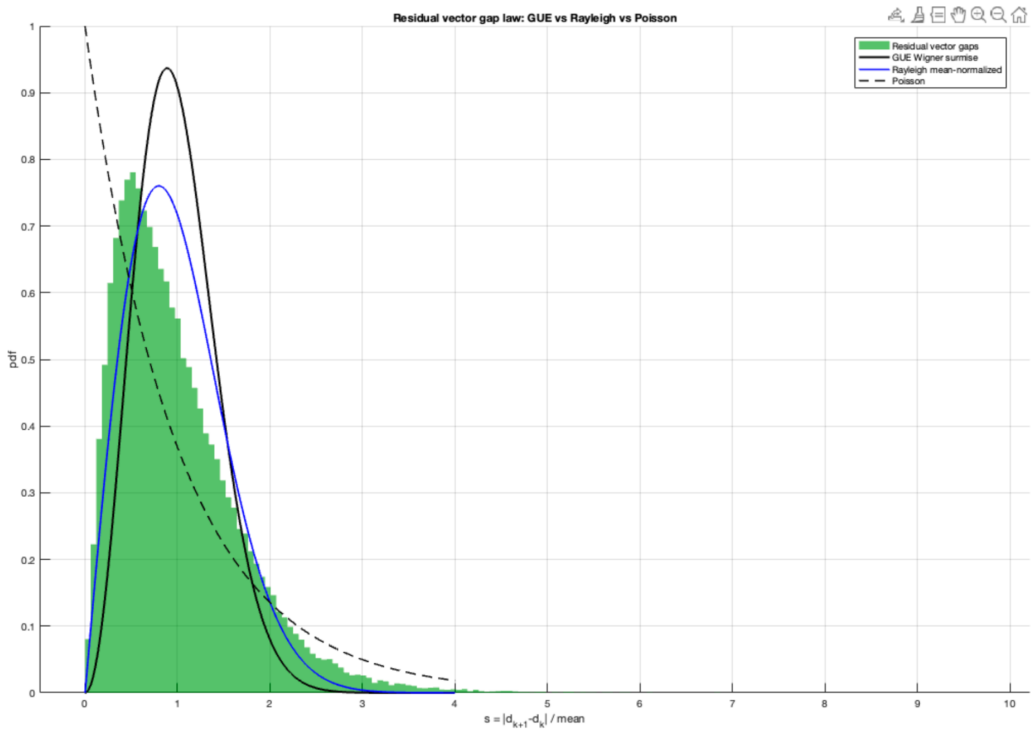


Figure 8: Nearest-neighbor residual vector gaps of the transformed function. After normalization by their mean, the residual gaps $|d_{k+1} - d_k|$ are compared with the GUE Wigner surmise, Poisson spacing, and the mean-normalized Rayleigh law.

Experiment 13.3. For $N = 100000$, the mean raw residual vector gap is

$$\mathrm{mean}_k |d_{k+1} - d_k| = 0.2569, \tag{13.5}$$

and the median normalized residual vector gap is

$$\mathrm{median}(s^{\mathrm{diffvec}}) = 0.8493. \tag{13.6}$$

The Kolmogorov–Smirnov distances are

$$\mathrm{KS}_{\mathrm{GUE}} = 0.2752, \tag{13.7}$$
$$\mathrm{KS}_{\mathrm{Rayleigh}} = 0.08390, \tag{13.8}$$
$$\mathrm{KS}_{\mathrm{Poisson}} = 0.1555. \tag{13.9}$$

The near-zero CDF exponent, fitted from

$$P(s \le \varepsilon) \sim \varepsilon^{\alpha}, \tag{13.10}$$

is

$$\alpha = 1.963. \tag{13.11}$$

Remark 13.4. The value (13.11) is close to the Rayleigh CDF exponent 2, not to the GUE nearest-neighbor exponent 3, and not to the Poisson exponent 1. Thus the one-step residual vector gaps are Rayleigh-like, not GUE-like.

Heuristic 13.5. This is a heuristic interpretation. The Rayleigh law and the GUE spacing law describe different statistics. GUE spacing tests level repulsion and pair correlations in the zero ordinates themselves. The quantity $|d_{k+1} - d_k|$ is instead a one-step modulus of a complex residual phase difference after the logarithmic proper-time transform. If the local real and imaginary components of this transformed residual are approximately centered Gaussian components, as one expects from a Selberg-scale phase fluctuation passed through a smooth complex background, then its modulus is Rayleigh-like. Thus Rayleigh behaviour here does not contradict the Montgomery–Odlyzko GUE picture. It says that the one-step residual is locally planar, while the window statistics test ordered block accumulation.

Heuristic 13.6. This is a heuristic interpretation. The residual difference $d_{k+1} - d_k$ is a complex planar quantity. If its real and imaginary components behave approximately as centered Gaussian components at the one-step level, then its modulus is expected to follow a Rayleigh-type radial law. Experiment 13.3 is consistent with this interpretation.

14. Optional scale-clock interpretation

Heuristic 14.1. This is only a heuristic reparametrization of the clock. If one introduces

$$u = \log k, \tag{14.1}$$

then the logarithmic proper time becomes

$$\tau_k = \pi \frac{\log u}{u}. \tag{14.2}$$

One may formally call this a scale-dependent clock or an effective redshift factor. No result in this article is derived from gravitational field equations, and no physical spacetime metric is proposed.

Warning 14.2. The scale-clock language is optional. Its only role is to make clear that the observable evolves different spectral levels with different effective times. The numerical claims of the article are the definitions of the transform, the comparison with the RVM control, and the finite window and gap diagnostics.

15. Finite computational observation and open tests

Experiment 15.1. For the first $N = 2000000$ ordinates in the finite computation, the residual window excess values in Experiment 10.5 imply the finite observations

$$\max_k R^{\mathrm{diff}}_{500}(k) = 8.4750, \tag{15.1}$$

$$\max_k R^{\mathrm{diff}}_{1000}(k) = 11.8480, \tag{15.2}$$

$$\max_k R^{\mathrm{diff}}_{2000}(k) = 15.8891. \tag{15.3}$$

These are finite output values, not a theorem about all heights.

Question 15.2. What is the distribution of $\max_k R^{\mathrm{diff}}_W(k)$ under null models that preserve the local size of δE_k but destroy the ordered phase structure? This question is necessary for deciding whether the observed window maxima represent more than the expected extreme values of a scanned correlated process.

Question 15.3. The computation up to $N = 2000000$ shows a stabilized upper-tube regime after the early transitions. Do further phase-block reorganizations occur at larger heights, producing new arms, tubes, or relocations of the cumulative geometry?

16. Counting-coordinate response profiles

The logarithmic proper-time observable studied in the preceding sections uses the ordinate coordinate γ_k. This section records a separate construction involving the real coordinate σ. Its role is different from the phase-walk observable: it is a signed real-coordinate response built around the critical coordinate $1/2$. It is defined without assuming RH. If an actual zero is supplied with real part $\sigma_k \neq 1/2$, the associated signed weight records that displacement from the critical line. If RH holds, the supplied real-coordinate displacement is zero for every nontrivial zero and the signed weight vanishes term by term in the natural limiting sense.

Remark 16.1. The parameter β in $D_X(\beta)$ is a supplied real coordinate. Therefore a plot of $D_X(\beta)$ as β varies should be read as a response profile of the construct, not as a statement that zeros exist at those externally chosen real parts. This is the correct scope of the object. The construct excludes, by definition, any off-critical contribution unless an off-critical real coordinate is supplied. Its potential value is as a formal starting point for analytic refinements involving the real coordinate, not as a standalone numerical search procedure.

16.1. Completed zeta function and local normalization

Definition 16.2. The completed zeta function is

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma\left(\frac{s}{2}\right)\zeta(s), \tag{16.1}$$

and satisfies the functional equation

$$\xi(s) = \xi(1-s). \tag{16.2}$$

Remark 16.3. For real σ and γ, the functional equation and the reality of ξ on the real axis give

$$|\xi(\sigma + i\gamma)| = |\xi(1-\sigma+i\gamma)|. \tag{16.3}$$

This is a standard symmetry of the completed zeta function, not a detector by itself.

Definition 16.4. Write a nontrivial zero as

$$\rho_k = \sigma_k + i\gamma_k, \tag{16.4}$$

ordered by nondecreasing positive ordinate and counted with multiplicity. Its real displacement from the critical line is

$$\delta_k = \sigma_k - \frac{1}{2}. \tag{16.5}$$

Definition 16.5. Let m_k be the multiplicity of the zero ρ_k of ξ, and let

$$a_k = \frac{\xi^{(m_k)}(\rho_k)}{m_k!}. \tag{16.6}$$

Thus $a_k \neq 0$, and locally

$$\xi(s) = a_k(s-\rho_k)^{m_k} + O\left((s-\rho_k)^{m_k+1}\right). \tag{16.7}$$

For a supplied real displacement $\delta \neq 0$, define the multiplicity-normalized coefficient

$$C_k(\delta) = \left|\frac{\xi(\rho_k+\delta)}{a_k\delta^{m_k}}\right|^{-2}, \tag{16.8}$$

whenever the expression is finite and nonzero. This coefficient is positive by construction whenever it is defined. In the simple-zero case $m_k = 1$, $a_k = \xi'(\rho_k)$, so (16.8) reduces to the usual first-derivative normalization.

16.2. Counting coordinate and response profile

Definition 16.6. Let

$$N(T) = \#\{k : 0 < \gamma_k \leq T\} \tag{16.9}$$

be the zero-counting function, with zeros counted according to the ordering convention above. At the ordinate γ_k, the finite counting coordinate is

$$u_k = \log N(\gamma_k) = \log k. \tag{16.10}$$

Definition 16.7. For $X > 0$ and a supplied real coordinate β, set

$$\delta = \beta - \frac{1}{2} \tag{16.11}$$

and define the response profile

$$D_X(\beta) = \sum_{\gamma_k \leq X} C_k(\delta)\left(N(\gamma_k)^{2\delta} - N(\gamma_k)^{-2\delta}\right). \tag{16.12}$$

Equivalently, using $N(\gamma_k) = k$,

$$D_X(\beta) = \sum_{k \leq N(X)} C_k(\delta)\left(k^{2\delta} - k^{-2\delta}\right). \tag{16.13}$$

Remark 16.8. Since $u_k = \log k$, each signed weight difference is

$$k^{2\delta} - k^{-2\delta} = 2\sinh(2\delta u_k). \tag{16.14}$$

Consequently, for fixed externally supplied δ, the sign of each defined nonzero summand is the sign of δ. This sign property is algebraic and contains no information about whether a zero actually has real part β.

Remark 16.9. If β is chosen externally, for example $\beta = 0.9$, the sign of the summands is dictated by the supplied displacement $\beta - 1/2$. Such a calculation describes the response of the formal real-coordinate profile at that supplied coordinate. It is different from evaluating the construct on an actual zero with known real part σ_k. The distinction is essential: an externally supplied β probes the profile; an actual σ_k records the real-coordinate displacement of a supplied zero.

16.3. Critical involution and the role of the actual real parts

Remark 16.10. Under the critical involution $s \mapsto 1 - s$, the real displacement changes sign,

$$\delta \longmapsto -\delta, \tag{16.15}$$

while the counting coordinate remains the positive coordinate

$$u_k = \log k > 0. \tag{16.16}$$

Thus the involution exchanges

$$e^{2\delta u_k} \longleftrightarrow e^{-2\delta u_k} \tag{16.17}$$

by changing δ, not by replacing the actual coordinate u_k with $-u_k$.

Warning 16.11. It is not correct to identify the actual counting coordinate with a transformation $u_k \mapsto -u_k$. The equality $u_k = -u_k$ would force $u_k = 0$, which is not the case for $k > 1$. The only valid exchange is the weight exchange in (16.17) at fixed positive u_k.

Definition 16.12. If a zero $\rho_k = \sigma_k + i\gamma_k$ is supplied with its actual real coordinate, define its actual real displacement by (16.5). The associated per-zero signed response is defined by

$$\mathcal{D}_k = 0 \qquad \text{if } \delta_k = 0, \tag{16.18}$$

and by

$$\mathcal{D}_k = C_k(\delta_k)\left(k^{2\delta_k} - k^{-2\delta_k}\right) \qquad \text{if } \delta_k \neq 0, \tag{16.19}$$

whenever the coefficient is defined. This quantity uses the actual σ_k, not an externally chosen β.

Remark 16.13. If RH holds, then $\delta_k = 0$ for every nontrivial zero and $\mathcal{D}_k = 0$ term by term by (16.18). If a zero is supplied with $\sigma_k \neq 1/2$, then $\delta_k \neq 0$ and (16.19) gives a signed response whose sign records the side of the critical line on which that supplied zero lies. Thus the construct is built without assuming RH, and it would respond immediately to an off-critical real coordinate if such a zero were supplied.

Remark 16.14. The article therefore contains two complementary coordinates. The logarithmic proper-time observable uses γ_k and reveals ordered phase geometry in the ordinate direction. The real-coordinate response profile uses σ_k as a supplied coordinate and records displacement from the critical line. In its present form this is a bookkeeping construct, not a complete analytic theory; its relevance is that it gives a clean starting point for possible refinements in which the real-coordinate response is connected to deeper analytic structure of ξ.

17. Conclusion

We have defined and analyzed a logarithmic proper-time spectral observable for zeta-zero ordinates, compared it with the RVM semiclassical background, and demonstrated its structural properties on the first 2,001,052 Odlyzko zero ordinates. A separate short null-test visualization uses $N = 800$ terms only for graphical readability and is not the scale of the main computation.

Main findings: The one-step residual vector gaps follow a Rayleigh law ($\mathrm{KS}_{\mathrm{Rayleigh}} = 0.084$) rather than GUE ($\mathrm{KS}_{\mathrm{GUE}} = 0.275$). The coherent signal emerges at the level of ordered window amplitudes, with residual window excess reaching factors of 8–16 above the random-walk scale for $W = 500$–2000.

Mechanism. The observable's sensitivity to ordered phase fluctuations is rooted in the spectral defect $\delta E_k = \gamma_k - \gamma_k^{\mathrm{RVM}}$, which equals the classical oscillatory term $S(T)$ in the Riemann–von Mangoldt formula (Remark 10.11). By Selberg's central limit theorem, $S(\gamma_k)$ has variance $\sim (1/2\pi^2)\log\log\gamma_k$, yielding δE_k of order $\sigma_\delta \approx 0.4$. The logarithmic proper-time weight $\tau_k = \pi \log\log k/\log k$ then amplifies these known phase fluctuations through quasi-resonant summation. The null experiment shows a sharper distinction: independent Gaussian perturbations of the correct scale over-excite the logarithmic transform, whereas

the true zeta sequence gives a controlled intermediate response. Thus the logarithmic clock is sensitive to Selberg-scale fluctuations, but the observed spiral and window morphology require the ordered correlations present in the zeta phase defect.

Clarification on GUE. The GUE Wigner surmise describes level correlations between distant zeros of $\zeta(s)$—a global spectral property. The Rayleigh law observed here describes one-step differences of a residual function under nonlinear phase projection—a local planar property. These are complementary statistics of different objects and do not contradict each other. Both hold simultaneously.

Scope and generality. The construction is not limited to zeta. Any ordered quantum chaotic spectrum, random-matrix spectrum, or spectral model with a smooth background and a level-dependent clock can be analyzed by the same method. The observable reveals phase-interference structure that local spacing statistics discard because they ignore ordering.

Two-coordinate bookkeeping. Section 16 adds a complementary real-coordinate construct using the abscissa σ_k. It is defined independently of RH: on the critical line its signed response vanishes, while a supplied zero with $\sigma_k \neq 1/2$ gives a nonzero signed response. In its present form it is a modest formal device, not a complete analytic theory, but it supplies a clean point of departure for possible refinements involving the real coordinate of zeros. This complements the ordinate-based phase geometry studied in the rest of the paper.

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